Research in CompMath **Name:** *Mary Kim*
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1. (a) Factor $2^{15} - 1 = 32,767$ into a product of two smaller positive integers.
   $2^{15} - 1 \;=\; (2^5 - 1)\left(1 + 2^b + 2^{2b} + ... + 2^{(a-1)b}\right)$
   set $b = 5, a = 3$
   $= \; (2^5 - 1)\left(1 + 2^5 + 2^{2(5)} + ... + 2^{(3-1)5}\right)$
   $= (2^5 - 1)\left(\sum_{n=0}^{2} 2^{5n}\right)$
   $= 31 \times 1057$

   (b) Find an integer x such that $1 < x < 2^{32,767} - 1$ and $2^{32,767} - 1$ is divisible by x.
   $2^{32767} - 1 \;=\; (2^{31} - 1)\left(1 + 2^{31} + 2^{62} + ... + 2^{(1056)(31)}\right)$
   This makes $2^{32767} - 1$ a factor of $2^{17} - 1$, making x = $2^{31} - 1 = 2,147,483,647$.

2. Make some conjectures about the values of n for which $n - 1$ is prime or the values of n for which $3n - 2n$ is prime.

   For all integers $n > 1$, $3^n - 1$ is even and therefore not prime.

   I noticed a pattern that if $3^n - 2^n$ is prime, then n must be prime.

Table 1: Primality Table for n, $3^n - 1$, and $3^n - 2^n$

n	n Prime?	$3^n - 1$	$3^n - 1$ Prime?	$3^n - 2^n$	$3^n - 2^n$ Prime?
2	yes	8	no	5	yes
3	yes	26	no	19	yes
4	no	80	no	65	no
5	yes	242	no	211	yes
6	no	728	no	665	no
7	yes	2186	no	2059	no
8	no	6560	no	6305	no
9	no	19682	no	19171	no
10	no	59048	no	58625	no
11	yes	177146	no	175099	no
12	no	531440	no	527345	no
13	yes	1594322	no	1586131	no
14	no	4782968	no	4766585	no
15	no	14348906	no	14316139	no
16	no	43046720	no	42981185	no
17	yes	129140162	no	129009091	yes

3. The proof of Theorem 3 gives a method for finding a prime number different from any in a given list of prime numbers.

(a) Use this method to find a prime different from 2, 3, 5, and 7.
According to Theorem 3, there are infinitely many prime numbers.
If $m = p_1 p_2 \cdots p_n + 1$, then m is a prime number.
$m = 2 \cdot 3 \cdot 5 \cdot 7 + 1 = 211$

(b) Use this method to find a prime different from 2, 5, and 11.
With the same formula, you get
$m = 2 \cdot 5 \cdot 11 + 1 = 111$.

4. Find five consecutive integers that are not prime.

The five consecutive integers 24, 25, 26, 27, and 28 are not prime.

I figured this out by counting up; I don't necessarily have a proof for this.

5. Use the table in Figure I.1 and the discussion on p. 5 to find two more perfect numbers.

(a) If $2^n - 1$ is prime, then $2^{n-1}(2^n - 1)$ is a perfect number.
For $n = 5$, $2^5 - 1 = 31$ is prime, so $2^4(2^5 - 1) = \mathbf{496}$ is a perfect number.
For $n = 7$, $2^7 - 1 = 127$ is prime, so $2^6(2^7 - 1) = \mathbf{8128}$ is a perfect number.

6. The sequence 3, 5, 7 is a list of three prime numbers such that each pair of adjacent numbers in the list differ by two. Are there any more such "triplet primes"?

No, because the prime numbers become more sparse as they get larger. For example, the next prime number after 7 is 11, which is a difference of 4 from 7. The next prime number would be 13, which is a difference of 2 from 11, but the next prime number after that is 17, which is a difference of 4 from 13. This pattern continues, and there are no more triplet primes.

7. A pair of distinct positive integers (m, n) is called amicable if the sum of all positive integers smaller than n that divide n is m, and the sum of all positive integers smaller than m that divide m is n. Show that (220, 284) is amicable.

The positive integers smaller than 284 that divide 284 are 1, 2, 4, 71, and 142. The sum of these numbers is: 220.

Similarily, the positive integers smaller than 220 that divide 220 are 1, 2, 4, 5, 10, 11, 20, 22, 44, 55, and 110. The sum of these numbers is: 284.